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Detecting Stealthy False Data Injection Using Machine Learning in Smart Grid

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***Abstract*—Aging power industries together with increase in the demand from industrial and residential customers are the main incentive for policy makers to define a road map to the next generation power system called smart grid. In smart grid, the overall monitoring costs will be decreased but at the same time, the risk of cyber attacks might be increased. Recently a new type of attacks (called the stealth attack) has been introduced, which cannot be detected by the traditional bad data detection using state estimation. In this paper, we show how normal operations of power networks can be statistically distinguished from the case under stealthy attacks. We propose two machine learning based techniques for stealthy attack detection. The first method utilizes the supervised learning over labeled data and trains a distributed support vector machine. The design of the distributed SVM is based on the Alternating Direction Method of Multipliers, which offers provable optimality and convergence rate. The second method requires no training data and detects deviation in measurements. In both methods, principle component analysis is used to reduce the dimensionality of the data to be processed, which leads to lower computation complexities. The results of the proposed detection methods on the IEEE standard test systems demonstrate the effectiveness of both schemes.**

Keywords: power system state estimation, bad data detection, anomaly detection, support vector machines

I. Introduction

Recently population growth on one and increasing consumerism in many societies on the other hand, has created major challenges for many industries such as electric industry. Facing these challenges need profound changes in traditional power systems, and should consider the improvement in operational efficiencies and environmental compliances. Transition to smart grid opens enormous research opportunities such as, 1) improving the quality of monitoring with utilizing advanced communication infrastructure [1]–[3]; 2) integration of new types of energy resources such as wind and solar energies [4]–[6].

Conventionally electricity is not storable in vast quantity, and as a result the generation should follow the consumption of electricity closely. Any mismatch of generation and consumption will deviate the electrical quantities such as frequency and voltage level in the power grid. Thus, control centers need to closely monitor the power network to make sure that the operation of the power system is safe and reliable. State estimation is an efficient way of online monitoring of states in the power networks. To estimate the states of network, the measured values of active power should be collected distributedly to a central state estimator using

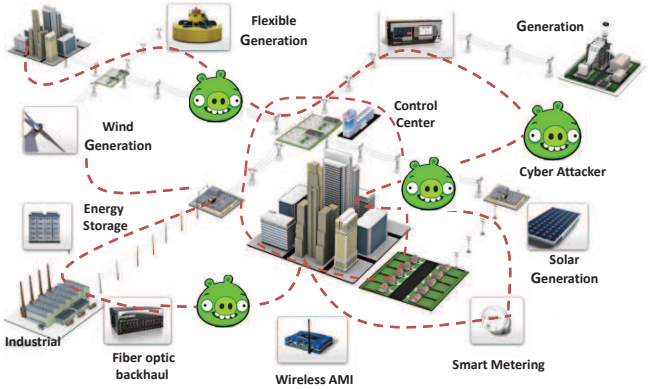

Fig. 1. Cyber threat in smart grid

communication links. These measurements are collected via sensors installed in the transmission network (mainly on substations), and consist of transmitted active power and injected active power measurements. The estimated states will be the base of corrective actions that the control center considers to keep the operation of power grid in a safe mode.

Communication infrastructure links connects the critical power facilities together, but at the same time it increases the cyber security challenges. Communication links used in state estimation can have the potential cyber attack risk. Different incentives may initiate the cyber attack in state estimators such as 1) making financial benefits 2) creating technical problems to the grid such as blackouts 3) a combination of financial benefits and technical problems. Analyzing the vulnerabilities in the state estimation (and more generally in smart grid) gives the chance to control centers to improve the safety of operation through considering suitable countermeasures.

Unlike other communication networks, practically the information exchange in smart grid is more likely in one direction. For example, sending information from measurements to the control center is usually in one direction (from measurements to the control center). This characteristic gives priority to the studying of malicious user behaviors [7]–[12] versus studying selfish behaviors [14]–[16] of users in smart grid. Stealth attack is an example of malicious behavior that in this paper we formulate and solve its detection based

on signal processing and machine learning techniques. This type of attack bypasses the conventional bad data detection methods and can create serious problems such as power system outages[1]. While different methods have been used for bad data detection in power system [7]–[13], to the best knowledge of the authors this is the first paper that uses machine learning methods to detect stealthy false data injection. We devise two machine learning based techniques for stealthy attack detection. The first method utilizes supervised learning over labeled data and trains a support vector machine. The second method requires no training data and detects the deviation in measurements. In both methods, we first use principle component analysis to project the data to a low dimensional space. The key observation that motivates our solution approach is that normal data and tampered data (due to attacks) tend to be separated in the projected space. This is because the normal data are governed by physical laws such as Kirchhoff's law, while the tampered data are not. Using PCA also reduces the dimensionality of the data to be processed and thus leads to lower computation complexities. Simulation results demonstrate the effectiveness of the machine learning methods in the separation of the attacked and safe operation modes.

II. Literature Survey

Security of smart grid is surveyed in [18] and [19]. [18] reviews cyber security for different parts of smart grid, such as Process Control System (PCS), smart meter, power system state estimation, and smart grid communication protocol. [19] surveys malicious attacks in three different categories based on the smart grid security objectives:

- *Attacks targeting availability:* Attackers try to delay, block or corrupt the communication in the smart grid (also called as denial-of-service attacks) [20], [21].
- *Attacks targeting integrity:* Attacker attempts to illegally disrupt the data exchange [7]–[12].
- *Attacks targeting confidentiality:* Attacker tries to get unauthorized information from network resources [22]–[24].

Challenges and opportunities of utilizing power networks with communication networks are described in [1]–[3]. False data may be due to unintended measurement abnormalities, topology errors, or injection by malicious attacks. An undetectable attack (stealth attack) is introduced in [7] where the authors show that how this type of false data passes the Bad Data Detector (BDD) in the control center. [8] and [9] analyze the economical effect of false data injection on the transmission line congestion. [10] shows that even if the structure of the power network is unknown, the attacker can first infer the structure and then use the Independent Component Analysis (ICA) to launch the stealth attack. In [12] authors define a security measure to quantify the hardness of performing attacks, and describes an algorithm to compute this hardness. [13] uses low rank structure of power grid as well as sparse nature of observable malicious attacks and formulates the false data detection problem as low rank matrix recovery and completion problem.

The remainder of the paper is organized as follows. The system model is given in Section III. False data injection and the stealthy false data injection are studied in Section III. The proposed machine learning techniques for false data detection are described in Section IV. Numerical results are given in Section V, and Section VI concludes the paper. Some important notations are listed in Table I.

TABLE I
Notations

P_{ij}	Transmitted power from bus i to bus j
$\boldsymbol{\theta}$	$n \times 1$ vector of voltage angles
X_{ij}	Line reactance between bus i and bus j
$\mathbf{H}$	$m \times n$ Jacobian matrix
m	Number of measurements
n	Number of states (number of buses here)
$\mathbf{r}$	$m \times 1$ residue vector for BDD
γ	Residue threshold in BDD
$\boldsymbol{\Sigma}_e$	$m \times n$ covariance matrix of measurements' errors
$\mathbf{z}'$	$m \times 1$ attacked measurement vector
$\mathbf{Z_t}$	Measurement sets over m different time steps
$\mathbf{Z_{tr}}$	$m \times k$ reduced measurement matrix
$\boldsymbol{\omega}$	Support Vector Machine optimization parameter
$\mathbf{f^{(i)}}$	Similarity function for i^{th} sample
F_1	Metric for evaluating the performance of clustering algorithms
$\boldsymbol{\xi}$	Support Vector Machine optimization parameter
δ	Threshold for Anomaly Detection algorithm

III. System Model

In power systems, transmission lines are used to transfer generated power to consumers [25]. Theoretically, the transmitted complex power between bus i and bus j depends on the voltage difference between these two buses, and it is also a function of impedance between these buses. In general, transmission lines have high reactance over resistance (i.e. X/R ratio), and one can approximate the impedance of a transmission line with its reactance. The transmitted active power from bus i to bus j can be written as [26]:

$$P_{ij} = \frac{V_i V_j}{X_{ij}} \sin(\theta_i - \theta_j), \tag{1}$$

where V_i is the voltage magnitude, θ_i is the voltage phase angle in bus i, and X_{ij} is the reactance of transmission line between bus i and bus j. In DC power flow studies, it is usually assumed that voltage phase difference between two buses are small, and the amplitudes of voltages in buses are near to the unity. Therefore, further simplification gives a linear relation between voltage phase angles and lines reactance as [27]:

$$P_{ij} = \frac{\theta_i - \theta_j}{X_{ij}}. \tag{2}$$

The state-estimation problem is to estimate n phase angles θ_i, by observing m real-time measurements. In power flow

[1]This is indeed because of the inaccurate reactions of the control center in response to the faulty-feedback through faulty state estimation that it receives.

studies, the voltage phase angle (θ_i) of the reference bus is fixed and known, and thus only $n-1$ angles need to be estimated. We fine the state vector as $\boldsymbol{\theta} = [\theta_1, \ldots, \theta_n]^T$.

The control center observes vector $\mathbf{z}$ for m active power measurements. These measurements could be either transmitted active power from bus i to bus j (P_{ij}), or injected active power to bus i ($\sum P_{ij}$). The observation can be described as follows:

$$\mathbf{z} = \mathbf{P}(\boldsymbol{\theta}) + \mathbf{e}, \tag{3}$$

where $\mathbf{z} = [z_1, \cdots, z_m]^T$ is the vector of measured active power in transmission lines, $\mathbf{P}(\boldsymbol{\theta})$ is the nonlinear relation between measurement $\mathbf{z}$ and, state $\boldsymbol{\theta}$ that is the vector of n bus phase angles θ_i, and $\mathbf{e} = [e_1, \cdots, e_m]^T$ is the Gaussian measurement noise vector with covariant matrix $\boldsymbol{\Sigma}_e$.

Without loss of generality the Jacobian matrix $\mathbf{H} \in \mathbb{R}^m$ at $\theta = 0$ is,

$$\mathbf{H} = \frac{\partial \mathbf{P}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \mid_{\boldsymbol{\theta}=\mathbf{0}} . \tag{4}$$

If the phase differences ($\theta_i - \theta_j$) in (2) are small, the linear approximation model of (3) can be described as:

$$\mathbf{z} = \mathbf{H}\boldsymbol{\theta} + \mathbf{e}. \tag{5}$$

We focus on the problems of identifying and mitigating the impact of malicious cyber attacks on state estimation, as state estimation plays a key role of in connecting measurements collected via the communication network and controlling physical operations in smart grid. Next, we first study how to detect false data injection, and then explain the concept of stealth attacks.

A. Conventional Bad Data Detection

Given the power flow measurements $\mathbf{z}$, the estimated state vector $\hat{\boldsymbol{\theta}}$ can be computed as:

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T \boldsymbol{\Sigma}_e^{-1} \mathbf{H})^{-1} \mathbf{H}^T \boldsymbol{\Sigma}_e^{-1} \mathbf{z} = \mathbf{M}\mathbf{z}, \tag{6}$$

where $\mathbf{M} = \mathbf{H}(\mathbf{H}^T \boldsymbol{\Sigma}_e^{-1} \mathbf{H})^{-1} \mathbf{H}^T \boldsymbol{\Sigma}_e^{-1}$. Thus, the residue vector $\mathbf{r}$ can be computed as the difference between the measured quality and the calculated value from the estimated state: $\mathbf{r} = \mathbf{z} - \mathbf{H}\hat{\boldsymbol{\theta}}$. Therefore, the expected value and the covariance of the residual are:

$$E(\mathbf{r}) = \mathbf{0} \text{ and } cov(\mathbf{r}) = (\mathbf{I} - \mathbf{M})\boldsymbol{\Sigma}_e. \tag{7}$$

False data detection due to faulty sensors and topological errors can be performed using a BDD such as threshold test proposed in [30]. Therefore, the hypothesis of not being attacked is accepted if

$$\max_i |r_i| \leq \gamma, \tag{8}$$

where γ is the threshold and r_i is the component of $\mathbf{r}$.

B. Stealth (unobservable) attack

From the discussion of BDD, we observe that if the attacker has knowledge on topology $\mathbf{H}$, it can add $\delta\boldsymbol{\theta}$ to $\hat{\boldsymbol{\theta}}$ by changing the value of measurement data as,

$$\mathbf{z}' = \mathbf{H}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) + e. \tag{9}$$

In this case, θ and $\theta + \delta\theta$ are within the same affine set, thus $\max_i |r_i| \leq \gamma$ doesn't change and the hypothesis test fails in detecting the attacker. Indeed the control center believes that the true state is $\boldsymbol{\theta} + \delta\boldsymbol{\theta}$. This is called *stealth false data injection* [7].

IV. Machine Learning Based Bad Data Detection

In the previous section, we have showed that stealth bad data can pass the traditional BDD. In this section, we devise two machine learning based techniques to detect stealthy attacks. The main motivation for this approach is that normal data and tampered data (due to attacks) tend to be separated in certain projected space. When the class labels (normal vs. tampered) are given in the historical data, we can train a classifier to identify attacks. On the other hand, when labels are not given, we will apply anomaly detection to identify the outliers as potential attacks. In both schemes, one main challenge is the curse of dimensionally, namely, as the size of the power grid grows, the dimension of the measurement data grows rendering high computation complexity. We use principle component analysis (PCA) to first reduce the dimension of measurements, and then apply the proposed classification/detection techniques. PCA maps the data from the original domain to a new domain. Attacked data in the new domain are more handleable than the original domain because the data are not correlated anymore.

A. Pre-Processing Principle Component Analysis

Most of the practical systems such as power networks have complex structure, thus understanding their behaviour is very challenging in some cases. One interesting way of analyzing these behaviour is to first use redundant measurements in the network, and then use PCA to extract the interest dynamics of the system. PCA is a well known method, so we just briefly bring the concept and formulation here. Mathematically, PCA maps the data from n dimensional space to r dimensional space (12) where $r \leqslant n$. The data in the new domain have two important properties: 1) different dimensions of the data have no correlation anymore 2) the dimensions are ordered based on the importance of their information. The following equations are used to map the $m \times n$ measurement matrix $\mathbf{Z_t}$ to an $m \times r$ dimension matrix $\mathbf{Z_{tr}}$ [31],

$$\eth = \frac{1}{m} \times \mathbf{Z_t^T} \times \mathbf{Z_t}, \tag{10}$$

$$[\mathbf{U}, \mathbf{S}, \mathbf{V}] = svd(\eth), \tag{11}$$

$$\mathbf{Z_{tr}} = \mathbf{U}(:, 1:r)^T \times \mathbf{Z_t}, \tag{12}$$

where $\mathbf{Z_t} = [\mathbf{z}^{(1)}, \ldots, \mathbf{z}^{(m)}]^T$ is the matrix containing measurement sets over m different time steps. $\mathbf{S}$ is a diagonal matrix with nonnegative diagonal elements in a decreasing

order. Matrices $\mathbf{U}$ and $\mathbf{V}$ are unitary matrices that satisfy $\eth = \mathbf{USV}^T$. svd is a function for computing singular value decomposition. S_{ii} is the eigenvalue of the ith feature where the bigger S_{ii} is, the more information i^{th} feature has[2]. Indeed in many correlated systems (such as power grids), only first few components of S are significant. It is common to select the smallest value of r such that the following condition holds,

$$\frac{\sum_{i=1}^{r} S_{ii}}{\sum_{i=1}^{m} S_{ii}} \times 100 \geq \varepsilon, \tag{13}$$

namely $\varepsilon\%$ of variance is retained. After mapping the features to the low dimensional space in (12), the control center can use efficient machine learning techniques to determine the boundary between the normal and tampered data.

Toward this end, we can observe that PCA chooses the principle components that maximize the variance. This is a very important feature, which explains why with much less information (or number of dimensions) of the observations, the anomaly detection still performs well. PCA not only retains all the signal variations, but it also maximize the separation between normal and anomaly operation points. This property is precious, since it helps to increase the attack detection property and explains why we choose PCA to reduce the number of observations dimensions.

B. Detection Method 1: Anomaly Detection

In data mining, the data sets considerably different from the remainder of data are called outliers or anomalies. Different types of anomaly detection methods have been proposed such as the distance-based, model based, and statistical methods [29]. In this paper, we use the statistical based methods. We use metric $P(\mathbf{z})$ and a threshold δ, where $P(\mathbf{z})$ represents the statistical characteristics of the historical data. If $P(\mathbf{z}) \leq \delta$, then $\mathbf{z}$ statistically has low similarity to the remaining data. In this method, the hypothesis of anomaly is confirmed if $P(\mathbf{z}) \leq \delta$, and it is rejected if $P(\mathbf{z}) > \delta$. Threshold δ will be learnt by the historical data (Step 3. in Algorithm 1). Because of using historical data to learn δ, this method in some literature is called semi–supervised learning method. We use the multivariate Gaussian distribution probability density function as the metric $P(\mathbf{z})$,

$$\mathbf{P}(\mathbf{Z}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{n}{2}} |\boldsymbol{\Sigma}|^{0.5}} \exp\left[-\frac{1}{2}(\mathbf{Z} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{Z} - \boldsymbol{\mu})\right], \tag{14}$$

$$\boldsymbol{\mu} = \frac{1}{m}\sum_{i=1}^{m} \mathbf{Z}^{(i)}, \quad \boldsymbol{\Sigma} = \frac{1}{m}\sum_{i=1}^{m} (\mathbf{Z}^{(i)} - \boldsymbol{\mu})(\mathbf{Z}^{(i)} - \boldsymbol{\mu})^T,$$

where n is the number of features, m is the number of samples and $P_i(z_i)$ is the probability density function (PDF) of feature i. Each feature z_i follows a certain distribution that should be fitted based on the historical data. It is worth mentioning that the assumption of independency for $\mathbf{Z}$'s holds for Gaussian distributed features because of using PCA[3]. Algorithm 1 shows the basic procedure of the anomaly detection method.

In Step 1, historical data will be collected. These data will be gathered from the SCADA information in the control center. Collected historical data will be pre-processed and will be mapped to a low dimensional space in Step 2. This will decrease the computational complexity without sacrificing accuracy [31]. In Step 3, a Gaussian density function will be fitted to the pre-processed data ($\mathbf{P}(\mathbf{Z})$). If the new operational points (validation data set which have not been used in Step 3) statistically are not similar to the historical data , the value of $\mathbf{P}(\mathbf{Z_{val}})$ will be less than a threshold δ. Step 4 defines the best possible δ using the labeled historical data. New operating point will be tested in Step 5 to see how similar is it to the normal data.

Algorithm 1: Stealth false data detection using anomaly detection

1 Collect historical data from state estimator
$\mathbf{Z_t} = [\mathbf{z}^{(1)}, \ldots, \mathbf{z}^{(m)}]^T$,
2 Use PCA:
$\eth = \frac{1}{m} \times \mathbf{Z_t^T} \times \mathbf{Z_t}$;
$[\mathbf{U}, \mathbf{S}, \mathbf{V}] = svd(\eth)$;
$\mathbf{Z_{tr}} = \mathbf{U}(:, 1:k)^T \times \mathbf{Z_t}$;
3 Fit density function $\mathbf{P}(\mathbf{Z}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$ to the historical data $\mathbf{Z_t}$ using (14);
4 Choose the best δ;
$\delta^{best} = 0,\ F_1^{best} = 0,$
for $\delta = min\mathbf{P}(\mathbf{Z_{val}}) : St : max\mathbf{P}(\mathbf{Z_{val}})$ **do**

$$\mathbf{Y_{pred}} = \begin{cases} \mathbf{Y_{pred}}(i) = 1, & \forall i, \quad P(Z_{val})(i) \leq \delta, \\ \mathbf{Y_{pred}}(i) = 0, & \forall i, \quad P(Z_{val})(i) > \delta, \end{cases}$$

$f_p = sum(\mathbf{Y_{pred}} == 1 \ \&\ \mathbf{Y} == 0)$,
$t_p = sum(\mathbf{Y_{pred}} == 1 \ \&\ \mathbf{Y} == 1)$,
$f_n = sum(\mathbf{Y_{pred}} == 0 \ \&\ \mathbf{Y} == 1)$,
$t_n = sum(\mathbf{Y_{pred}} == 0 \ \&\ \mathbf{Y} == 0)$,

$$F_1 = 2\frac{P_r \times R_e}{P_r + R_e}, \text{ where} \begin{cases} P_r = t_p/(t_p + f_p), \\ R_e = t_p/(t_p + f_n), \end{cases}$$

if $F_1 > F_1^{best}$ **then**
$F_1^{best} \leftarrow F_1 \ \ \delta^{best} \leftarrow \delta$ exit
end
5 For the new operating point $\mathbf{Z^{new}}$

$$\text{If } P(\mathbf{Z^{new}}; \mu, \boldsymbol{\Sigma}) = \begin{cases} \leq \delta^{best} \dashrightarrow \mathbf{Z^{new}} \text{ is corrupted,} \\ > \delta^{best} \dashrightarrow \mathbf{Z^{new}} \text{ is normal.} \end{cases}$$

C. Detection Method 2: Support Vector Machines

In this subsection, a distributed Support Vector Machine (SVM) is proposed to detect stealth false data injections. Generally, a SVM constructs a hyperplane or a set of

[2]Statistically, direction with the highest variation is the most important direction because it can represent the best approximation of the data in lower dimensions (direction with highest variation, has the most important information among other directions inside the data).

[3]PCA transforms a set of (possibly) correlated data into the linearly uncorrelated data.

hyperplanes used for classification and regression. Given a labeled training set $\mathcal{S} = (\mathbf{x}_l, y_l), l = 1, \ldots, L$ of size L, and $y_l \in \{1, -1\}$. The SVM can be obtained by solving:

$$\min_{\boldsymbol{\omega},\xi,b} \quad \frac{1}{2}\boldsymbol{\omega}^\top\boldsymbol{\omega} + C\sum_{i=l}^{L}\xi_l, \tag{15}$$

$$\text{subject to} \quad y_i(\boldsymbol{\omega}^\top\phi(\mathbf{x}_l) + b) \geq 1 - \xi_l, \tag{16}$$

$$\xi_l \geq 0, l = 1, \ldots, L, \tag{17}$$

where $\phi(\mathbf{x}_l)$ is a nonlinear transformation maps $\mathbf{x}_l$ in a higher dimensional space. The slack variable ξ_l accounts for non-linearly separable training sets, and C is a tunable positive regularization parameter.

In order to obtain a distributed SVM, (15) can be rewritten as:

$$\min_{\boldsymbol{\omega}_i,\xi_i,b_i} \quad \frac{1}{2}\sum_{i=1}^{N}\boldsymbol{\omega}_i^\top\boldsymbol{\omega}_i + C\sum_{i=1}^{N}\sum_{l=1}^{L}\xi_{il}, \tag{18}$$

$$\text{subject to} \quad y_{il}(\boldsymbol{\omega}_i^\top\phi(\mathbf{x}_{il}) + b_i) \geq 1 - \xi_{il}, \tag{19}$$

$$\xi_{il} \geq 0, i = 1, \ldots, N; l = 1, \ldots, L, \tag{20}$$

where N is the number of groups that work together to train the SVM and $\boldsymbol{\omega}_i$ is the local optimization parameter for each group. By introducing a global variable $\mathbf{z}$, the (18) can be reformulated as:

$$\min_{\mathbf{z},\boldsymbol{\omega}_i,\xi_i,b_i} \quad \frac{1}{2}\mathbf{z}^\top\mathbf{z} + C\sum_{i=1}^{N}\sum_{l=1}^{L}\xi_{il}, \tag{21}$$

$$\text{subject to} \quad y_{il}(\boldsymbol{\omega}_i^\top\phi(\mathbf{x}_{il}) + b_i) \geq 1 - \xi_{il}, \tag{22}$$

$$\xi_{il} \geq 0, \mathbf{z} = \boldsymbol{\omega}_i, \tag{23}$$

$$i = 1, \ldots, N; l = 1, \ldots, L. \tag{24}$$

In order to solve (21) distributively, variables $\{\mathbf{z}, \boldsymbol{\omega}_i\}, i = 1, \ldots, N$ can be partitioned into two groups $\{\mathbf{z}\}$ and $\{\boldsymbol{\omega}_i\}, i = 1, \ldots, N$, and the Alternating Direction Method of Multipliers (ADMM) can be applied to solve it. Specifically, the scaled augmented Lagrangian function is expressed as:

$$\mathcal{L}\{\mathbf{z}, \boldsymbol{\omega}_i, \xi_i, \rho, \boldsymbol{\mu}_i\} = \frac{1}{2}\mathbf{z}^\top\mathbf{z} + C\sum_{i=1}^{N}\sum_{l=1}^{L}\xi_{il} + \frac{\rho}{2}\|\boldsymbol{\omega}_i - \mathbf{z} + \boldsymbol{\mu}_i\|_2^2, \tag{25}$$

where ρ is the step size and $\boldsymbol{\mu}_i$ is the scaled dual variable. At each iteration k, $\{\boldsymbol{\omega}_i\}$, $\{\mathbf{z}\}$ and $\boldsymbol{\mu}_i$ are updated as follows:

$$\boldsymbol{\omega}_i[k+1] = \arg\min_{\boldsymbol{\omega}_i,\xi_i,b_i} C\sum_{l=1}^{L}\xi_{il} + \frac{\rho}{2}\|\boldsymbol{\omega}_i - \mathbf{z}[k] + \boldsymbol{\mu}_i[k]\|_2^2, \tag{26}$$

$$\text{subject to} \quad y_{il}(\boldsymbol{\omega}_i^\top\phi(\mathbf{x}_{il}) + b_i) \geq 1 - \xi_{il}, \tag{27}$$

$$\xi_{il} \geq 0, l = 1, \ldots, L. \tag{28}$$

Note that the update process of $\boldsymbol{\omega}_i$ can be done locally at the i_{th} group. Furthermore, it involves fitting a SVM to the local data with an offset in the quadratic regularization term, which can be solved easily with existing software like LIBSVM [17]. The vector $\{\mathbf{z}\}$ is expressed as:

$$\mathbf{z}[k+1] = \arg\min_{\mathbf{z}} \frac{1}{2}\mathbf{z}^\top\mathbf{z} + \frac{\rho}{2}\|\boldsymbol{\omega}_i[k+1] - \mathbf{z} + \boldsymbol{\mu}_i[k]\|_2^2, \tag{29}$$

which can be solved analytically as:

$$z = \frac{N\rho}{\frac{1}{C} + N\rho}(\bar{\boldsymbol{\omega}}[k+1] + \bar{\mu}[k]), \tag{30}$$

where $\bar{\boldsymbol{\omega}} = \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{\omega}_i$ and $\bar{\mu} = \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{\mu}_i$. Finally we update the scaled dual variable $\boldsymbol{\mu}_i$ by:

$$\boldsymbol{\mu}_i[k+1] = \boldsymbol{\mu}_i[k] + \boldsymbol{\omega}_i[k+1] - \mathbf{z}[k+1]. \tag{31}$$

Algorithm 2: Stealth false data detection using distributed SVM

Input: Historical data from state estimator;
Initialize: $\{\mathbf{z}\}$, $\{\boldsymbol{\omega}_i\}$, $\boldsymbol{\mu}_i$, ρ, $k = 0$;
while not converge **do**
  $\{\boldsymbol{\omega}_i\}$-update, distributively at each computing node:
  $\boldsymbol{\omega}_i[k+1] = \arg\min_{\boldsymbol{\omega}_i,\xi_i,b_i} C\sum_{l=1}^{L}\xi_{il}$
  $+\frac{\rho}{2}\|\boldsymbol{\omega}_i - \mathbf{z}[k] + \boldsymbol{\mu}_i[k]\|_2^2$,
  subject to Constraints (27),(28).

  $\{\mathbf{z}\}$-update:
  $z = \frac{N\rho}{\frac{1}{C}+N\rho}(\bar{\boldsymbol{\omega}}[k+1] + \bar{\mu}[k])$
  $\boldsymbol{\mu}_i[k+1] = \boldsymbol{\mu}_i[k] + \boldsymbol{\omega}_i[k+1] - \mathbf{z}[k+1]$.
  Adjust penalty parameter ρ_i is necessary;
  $k = k + 1$;
end while
return $\{\mathbf{z}\}$, $\{\boldsymbol{\omega}_i\}$;
Output $\{\mathbf{z}\}$, $\{\boldsymbol{\omega}_i\}$;

Algorithm 2 describes the procedure to detect stealth false data injection using the distributed SVM method. In Step 1, historical data are prepared at each group. These data can be obtained from the control center, or collected locally by each group. The local and global optimization parameters are initialized in Step 2. Step 3 is consisted of two parts: local parameters updates and global parameter update. The local optimization parameter ω_i is obtained by solving the optimization problem (26) under the constraints (27) and (28). This can be implemented by existing software like LIBSVM [17]. The global optimization parameter is optimized by (30), in which $\bar{\boldsymbol{\omega}}$ and $\bar{\boldsymbol{\mu}}$ can be obtained by in-network processing through messages exchanged only among neighboring group. After the problem is solved and each group achieved the converged ω_i, local and global optimization parameters, ω_i and z, are returned in Step 4.

V. Numerical Results

In this section, we evaluate the effect of machine learning based techniques for detecting stealth attack in state estimation. We use the IEEE 118–bus test system. In order to simulate more realistic operation of the power system, we will use the stochastic loads in the network. Without loss

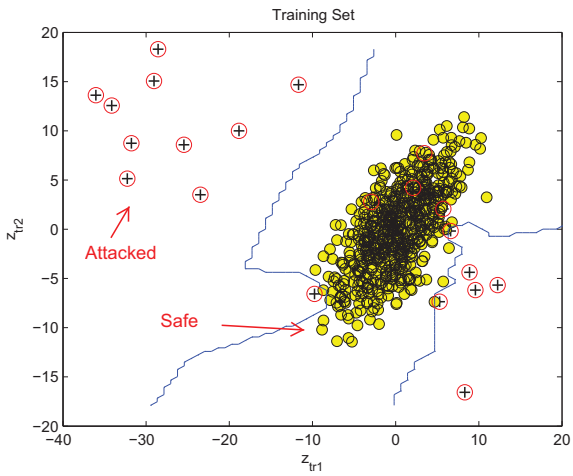

Fig. 2. Attacked and safe operating modes in $\mathbb{R}^2$ space

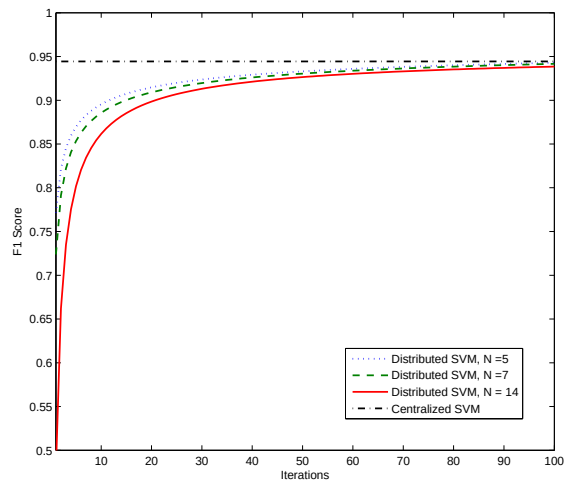

Fig. 3. Distributed SVM convergence

of generality, these loads are considered to follow a uniform distribution in the range of $[0.9 \times L_0 - 1.1 \times L_0]$, where L_0 is the base load. The mean value of load in a specific period of time is often considered as L_0. Here, we use MATPOWER [32] to simulate the operation of the power network. In these simulations, active power measurements are collected from each transmission line. Thus, in the 118-bus case study there are 304 measurement features (one feature per transmission line). These measurements will be considered as inputs to the proposed algorithms. Due to the random nature of the load, the measurement vector varies over the time. Using the Monte-Carlo simulation we have recorded measurement vector in 1000 different instances[4]. As previously discussed, the measurements data are highly correlated, and thus PCA is applied for dimension reduction. In the simulated data set, with only 2 principal components ($k = 2$), 99% of variance will be retained. Figure 2 shows a 2–dimensional plot of the principal components, where the circles mean in the normal state, and the plus signs mean in the attacked state. This figure demonstrates that stealth attack is separable in the control center.

A. Support Vector Machine

In this paper, we use the Gaussian kernel for the similarity function,

$$f_m^{(i)} = \exp(-\frac{\|\mathbf{z}_{tr}^{(i)} - \mathbf{l}^{(m)}\|^2}{2\sigma^2}), \tag{32}$$

where $\mathbf{l}^{(m)}$ is the mth landmark. The landmarks can be placed randomly in the historical data set space. σ is another parameter that can be used for changing the complexity of the decision boundary. Optimal choice for σ and C can improve the efficiency of SVM in detecting the attacked mode for cross validation set. In this paper, we have used 70% of the historical data as learning data set and tested the accuracy of the fitted decision boundary on the 30% of the remaining data sets called cross validation data sets. In order to define the optimal choice for σ and C, we vary both and choose the best σ and C, which correspond to the largest accuracy. In this paper, we use F_1 score as the measure of accuracy, i.e.,

$$F_1 = 2\frac{P_r \times R_e}{P_r + R_e}, \tag{33}$$

where P_r and R_e are called precision and recall, respectively, and they are calculated using the following equations,

$$P_r = \frac{\text{True Positive}}{\text{Predicted Positive}}, \; R_e = \frac{\text{True Positive}}{\text{Actual Positive}},$$

where true positive corresponds to the points that the algorithm detects as positive samples and they are indeed positive ones. Predictive positives are the points that the algorithm detects as positive points but it may have errors. Actual positive are all positive points in the data sets. $F1$ score is no greater than 1, and the bigger the value of $F1$, the more accurate the classifier in general.

The convergence performances are shown in Figure 3. It is shown that after a moderate number of iterations, the proposed distributed algorithm converges to the optimal values in different cases. From Figure 3, we can see that when the number of groups $N = 5$, the convergence rate is fastest. As the number of groups increases, the convergence rate slows down. This is mainly due to the increase of the communication overhead. Note that after very few iterations, the optimization result are very close to the optimal value, which means that the proposed algorithm are able to yield a good approximation to the optimal value in a short period of time.

To define the attacked and safe mode's boundary, we use a non-linear classifier with a Gaussian–kernel. Different values

[4]In the PJM (Pennsylvania, New Jersey, and Maryland) market, the control center collects the measured data in 1-minute time intervals and runs the Siemens state estimation program [33].

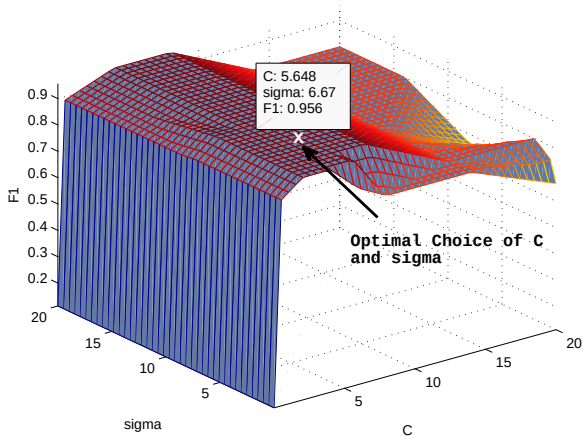

Fig. 4. Optimal Choice for C and σ

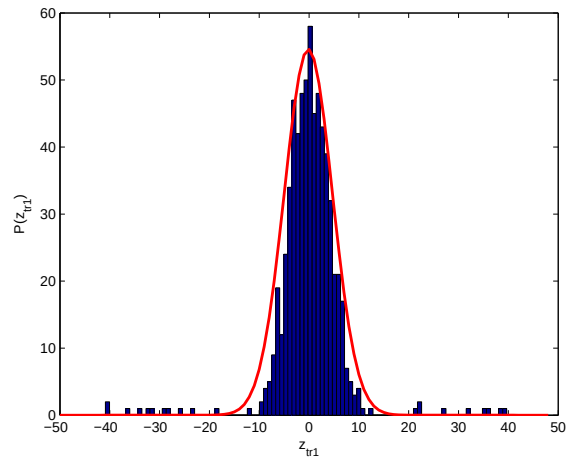

Fig. 5. Histogram representation of $\mathbf{z_{tr1}}$

of C and σ have different effects on the clustering efficiency, so we train SVM with different C and σ and compute $F1$ for the cross validation set. Following Algorithm 2, we define the best choice for C and σ. Figure 4 shows F_1 score for different values of C and σ. Efficiency of the learning algorithm can be improved by increasing the number of learning data. In order to analyze the effect of increasing the number of learning data on detection performances, it is often useful to plot a learning curve. Figure 9 compares the performance of anomaly and SVM methods. This figure shows that for fewer training data sets, anomaly detection performs better than SVM but with sufficient training data set, SVM outperforms anomaly detection. Plotted F_1 score is the average of F_1 which are obtained from 100 different data sets.

B. Anomaly detection

After applying PCA, the measurement data are mapped to points in $2D$. Figure 5 shows the histogram of the first feature. We fit a Gaussian probability density function to the features. Results of anomaly detection on practical problems show that fitting the Gaussian density function for other data sets (which does not follow the Gaussian distribution) does not change the clustering efficiency drastically. Following the procedure given in Algorithm 1, the anomaly points can be detected by applying a threshold δ. The sensitivity of detecting a point as anomaly, depends on the magnitude of threshold δ. For larger values of δ, the algorithm is more sensitive and flags anomaly for most of the operating points (Figure 6). For smaller values of δ, the algorithm is less sensitive and may lose detection of some of attacked points (Figure 7). In semi–supervised method, output labels are used to learn the best threshold (Figure 8).

VI. Conclusion

In this paper, we first collect the normal and stealthily attacked operating points in the state estimator. We use the collected data from active power flow measurements

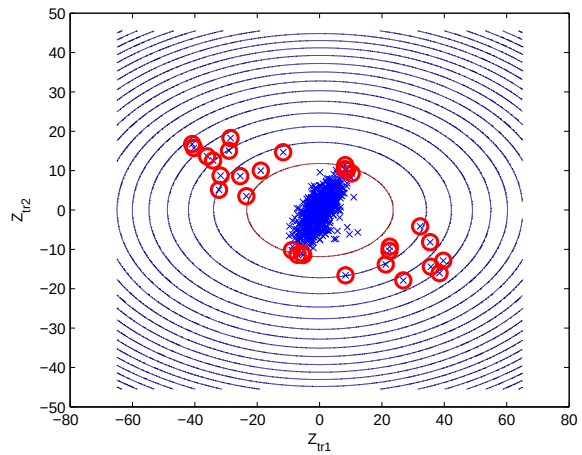

Fig. 6. Unsupervised–anomaly detection with large δ ($P(\mathbf{z}) < 2e-4$)

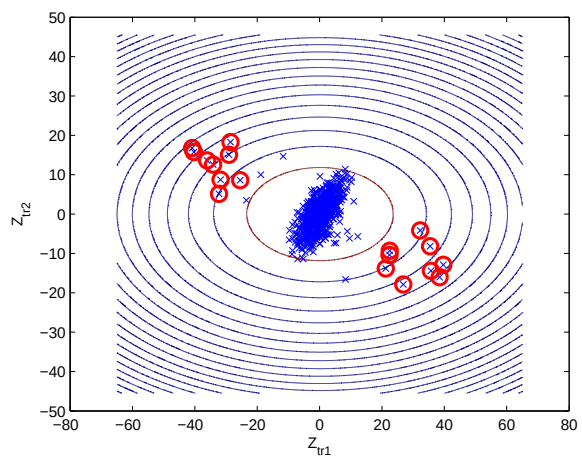

Fig. 7. Unsupervised–anomaly detection with small δ ($P(\mathbf{z}) < 2e-6$)

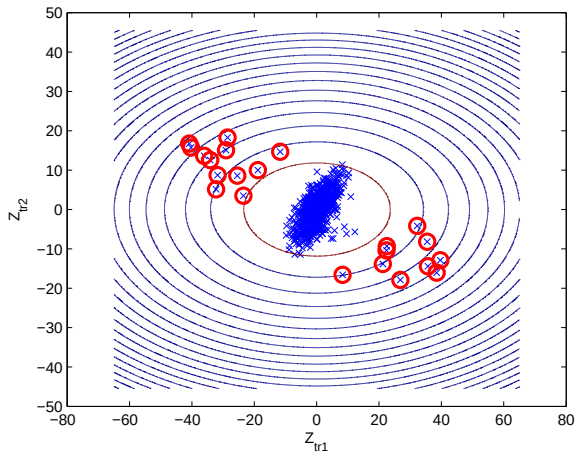

Fig. 8. Semi-supervised–anomaly detection with optimal δ ($P(\mathbf{z}) < 2.98e-5$)

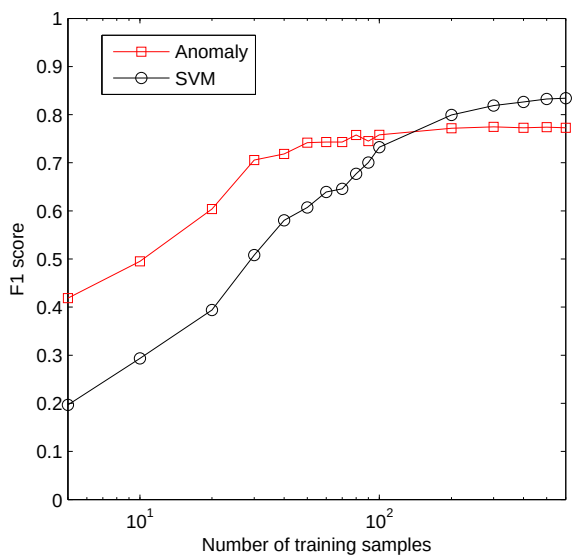

Fig. 9. Learning Curve of SVM and Anomaly Detection

in the network as the learning (historical) data. Projecting the historical data into a low dimensional space shows that normal measurement data are well separated from data under attack. This fact shows that the machine learning algorithms can be applied to detect the stealth false data injection in the state estimator. We use both supervised and unsupervised learning methods to distinguish the attacked and the safe operating modes. Numerical results shows the effectiveness of the proposed algorithms in detecting the stealth false data injection.

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